AI33-MPOPT · FOURTH PILLAR · 2026

Lawfulness

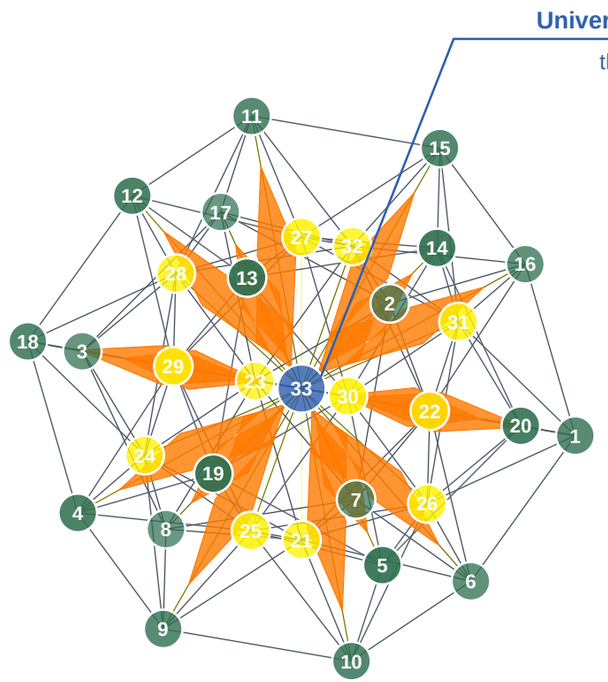
Four descriptions of one operator, measured as one structure — 97.4% in a single component, six signs out of six, in every configuration tested.

Rolando Rivero · Independent Researcher · AI33-MPOPT Framework

Companion to the Quantum-Gravity Bridge deck · for Zenodo

LAWFULNESS IN FOUR NUMBERS

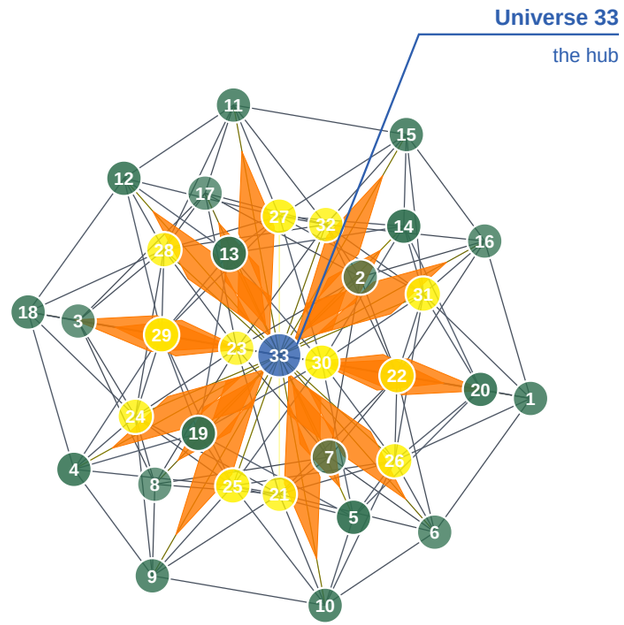
97.4%	6 / 6	30	0.9965
of the structure in one component	signs agree in every configuration	measured cells released as data	AUC · geometry recovered from information



20 curvilinears C1–C20
Hub · #33
12 inner · #21–32
20 outer · #1–20
32 throats to the hub
120 sibling links

GEOMETRY

The 33-multiverse, in plain terms



20 curvilinears C1–C20
Hub · #33
12 inner · #21–32
20 outer · #1–20
32 throats to the hub
120 sibling links

Every page after this one refers to the same fixed object. It is a **Platonic icosahedron** — the twenty-faced regular solid — used as an address system.

12 apexes icosahedron vertices — circular, direct line to the hub	the **dark-matter universes** — Universes 21–32
20 curvilinears icosahedron faces — three-sided regions between the apexes	the **dark-energy universes** — Universes 1–20, one seated on each curvilinear
32 siblings 12 + 20 together	the **32-multiverse** — the surrounding architecture, complete on its own
1 hub at the centre the 33rd	**Universe 33 — ours.** Not on the surface: at the geometric centre of the solid
32 throats centre to every sibling	the channels through which the 32-multiverse connects to the hub
120 sibling links across the surface	30 vertex–vertex + 60 vertex–face + 30 face–face — every count fixed by the solid itself

152 edges, 33 nodes, zero free parameters. The counts follow from the icosahedron — nothing is chosen or tuned. **The flow:** the 12 apexes reach the hub directly; the 20 dark-energy universes drive inward through the curvilinears, compacting as they pass and emerging in curvature motion. All 32 converge on one point at the centre. *Naming note: this framework calls the 12 vertices "apexes" and the 20 triangular faces "curvilinears"; in standard solid geometry those are the icosahedron's vertices and faces respectively. Both refer to the same object.*

FRAMEWORK

Lawfulness — what has to be true before physics

Before matter, before spacetime, before quantum states or curvature, one requirement has to hold: two descriptions of the same thing cannot contradict each other. That requirement is lawfulness. Not a force, not a field, not a geometry — a constraint.

WHY IT COMES FIRST

Quantum mechanics reads information. Gravity reads geometry. Neither contains the rule that binds them — so neither can judge the other. Lawfulness is the constraint both obey; quantum and gravity are outputs of it, not inputs to it.

HOW THIS FRAMEWORK MEASURES IT

Lawfulness stops being a word here: it is the agreement between an information field and a curvature field computed from the same operator. On the 32-multiverse structure that agreement runs 0.987 ± 0.003, and it holds on 13 of 17 graph families tested.

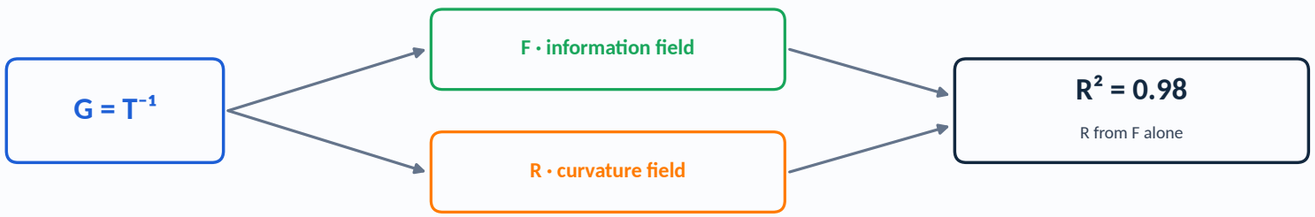
WHAT IT PREDICTS

If lawfulness is the deeper layer, then geometry should be recoverable from information alone — and it is: R is a near-direct function of F ($R^2 = 0.98$), with both derived from the single Green function $G = T^{-1}$.

WHERE IT STANDS

Lawfulness here is a measured consistency of the structure. Fifteen candidate explanations for the coupling were tested and eliminated, and the law is reported together with the metric it is measured in.

INFORMATION AND CURVATURE, ONE OPERATOR — AND HOW CLOSELY THEY AGREE



AGREEMENT 0.987 ± 0.003

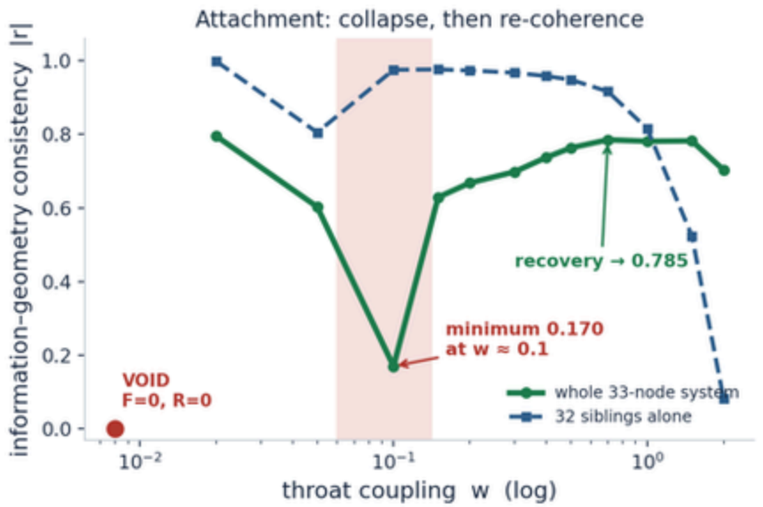
on the 32-multiverse structure

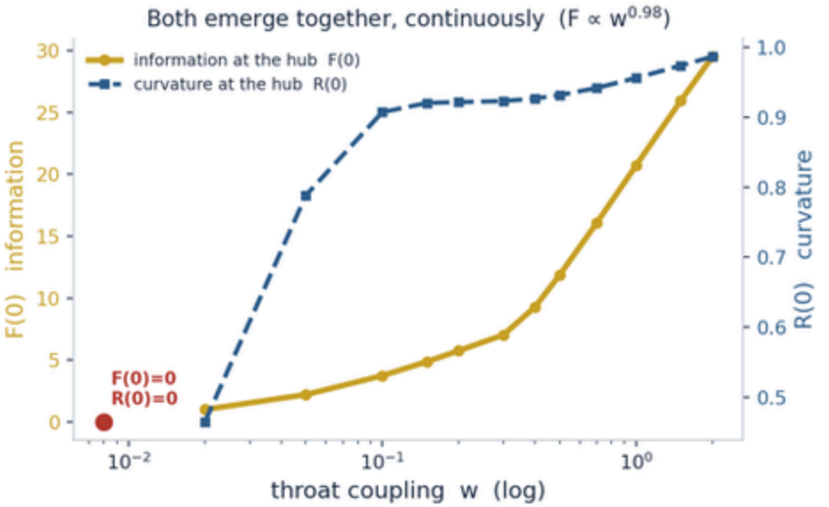
HOLDS ON 13 OF 17 GRAPH FAMILIES

The ordering this framework works from: **lawfulness** → **geometry** (the icosahedral 32-multiverse: 12 dark-matter vertices and 20 curvilinears (face-centres, dark energy)) → **inception** (Universe 33 forms at the geometric centre as the 32 throats connect) → **quantum and gravity** as two readings of that structure → **nature** as what happens inside one inception.

COMPUTED

The Void — lawfulness without relations





WHAT EXISTS IN THE VOID

Set the throat coupling to zero and the operator T becomes SINGULAR — it has no inverse, so there is no Green function and no propagator.

Every quantity vanishes exactly:
$F(0) = 0$ $R(0) = 0$
$G_{00} = 0$ $\max|G_{0j}| = 0$

No information. No curvature. No influence arriving. Not assumed — computed.

NO PROPAGATOR → NO 'HOW LONG'

Distance, travel time and causal order are all defined through the propagator. With no relations there is no propagator, so questions of duration have no answer in the void — not unknown, undefined.

THE SIBLINGS ARE ALREADY LAWFUL

While the hub sits at zero, the 32-sibling structure alone holds the law at 0.988. Lawfulness does not begin at inception; the hub enters a structure that is already coherent.

ATTACHMENT: COLLAPSE, THEN RE-COHERENCE

Weak coupling is disruptive: whole-system consistency falls to 0.170 at w ≈ 0.1. Above w ≈ 0.15 the structure re-coheres, rising to 0.785 by w ≈ 0.7. Inception has a threshold.

QUANTUM AND GRAVITY EMERGE TOGETHER

F(0) and R(0) leave zero at the same moment and rise together — F continuously, as w^0.98. Neither precedes the other; both appear as relations appear.

Interpretation, stated as such: the void is a lawful position with zero relations — where an information reading and a geometry reading are not yet distinguishable, because neither has anything to read.

Measure note: this page reports |r(F,R)| information–geometry consistency — the Quantum-Gravity Bridge deck's native quantity — not the one-factor share used elsewhere in this file. The same page appears as page 6 of that deck.

FOUNDATION

Lawfulness is not produced by the universe. The universe happens inside it.

1 LAWFULNESS

A structure of consistent relations. Not matter, not space, not time — only the constraint that descriptions of the same thing must agree with one another.

2 ARCHITECTURE

The 33-multiverse: a hub and thirty-two siblings joined by throats. What lawfulness looks like once it takes a shape.

3 INCEPTIONS

Not one. The outer-zone universes had theirs first. Universe 33 — ours — is a later event inside a structure that was already lawful.

4 QUANTUM AND GRAVITY

Present in every zone, expressed differently: different observables, different constants, different mathematics reading the same lawfulness from a different position in the structure.

THE CONSEQUENCE

Our Big Bang is not the beginning of law. It is one inception inside an already-lawful architecture. Quantum mechanics and gravity are not two things that must be joined — they are two readings of the lawfulness that was there first.

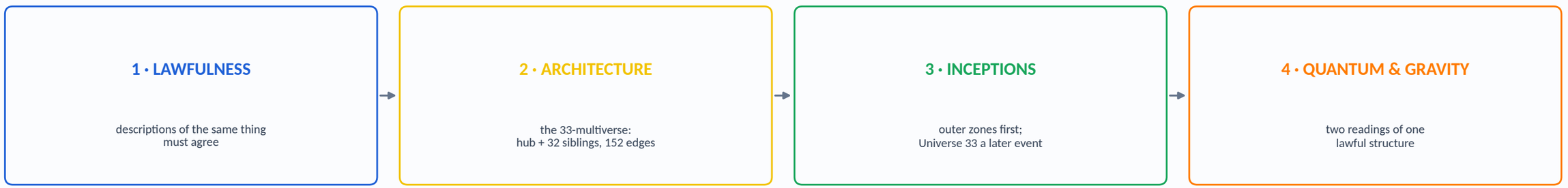
ABOUT THIS PAGE

This page states the framework's premise. Everything after it is the test of whether a structure built on this premise shows measurable consistency.

AND IT DOES

97.4% of the four-representation structure lies in a single component, with sign agreement six out of six in every configuration tested. The premise produces a system in which the descriptions are measurably one object.

THE LADDER: WHAT COMES FIRST, STEP BY STEP



FOUNDATION

Nothing is missing. Our position is partial.

A century of quantum gravity assumes the merger is something to be *performed* — take two theories and find the operation that joins them. Quantize gravity, or geometrize the quantum. Every major program does one or the other, and none has closed.

THE FRAMEWORK'S ANSWER

They were never two. What we call quantum mechanics and what we call gravity are two readings of one lawful structure, taken from inside a single late inception. The bridge is not missing — the vantage point is partial.

This is a reframe of the same structure, with the mechanism to be tested next. Einstein did not join gravity and acceleration. He observed that they had never been separate, and the field reorganised around that observation.

WHAT THIS FRAMEWORK DOES

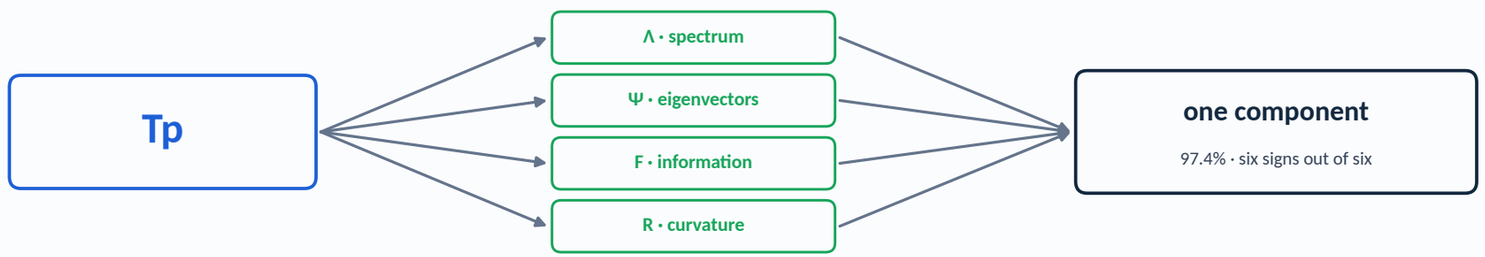
There is no merger operation anywhere in this construction. Nothing is joined, bridged, or unified by hand. One operator is built, four descriptions are taken from it, and the agreement between them is measured.

The result — 97.4% in one component, six signs out of six — is what already merged looks like when you measure it rather than attempt it.

WHAT IT ESTABLISHES

What it establishes is narrower and checkable: a system built on that premise produces, without being asked to, the unity that a century of merger attempts has been looking for.

ONE OPERATOR, FOUR DESCRIPTIONS — NOTHING IS JOINED BY HAND



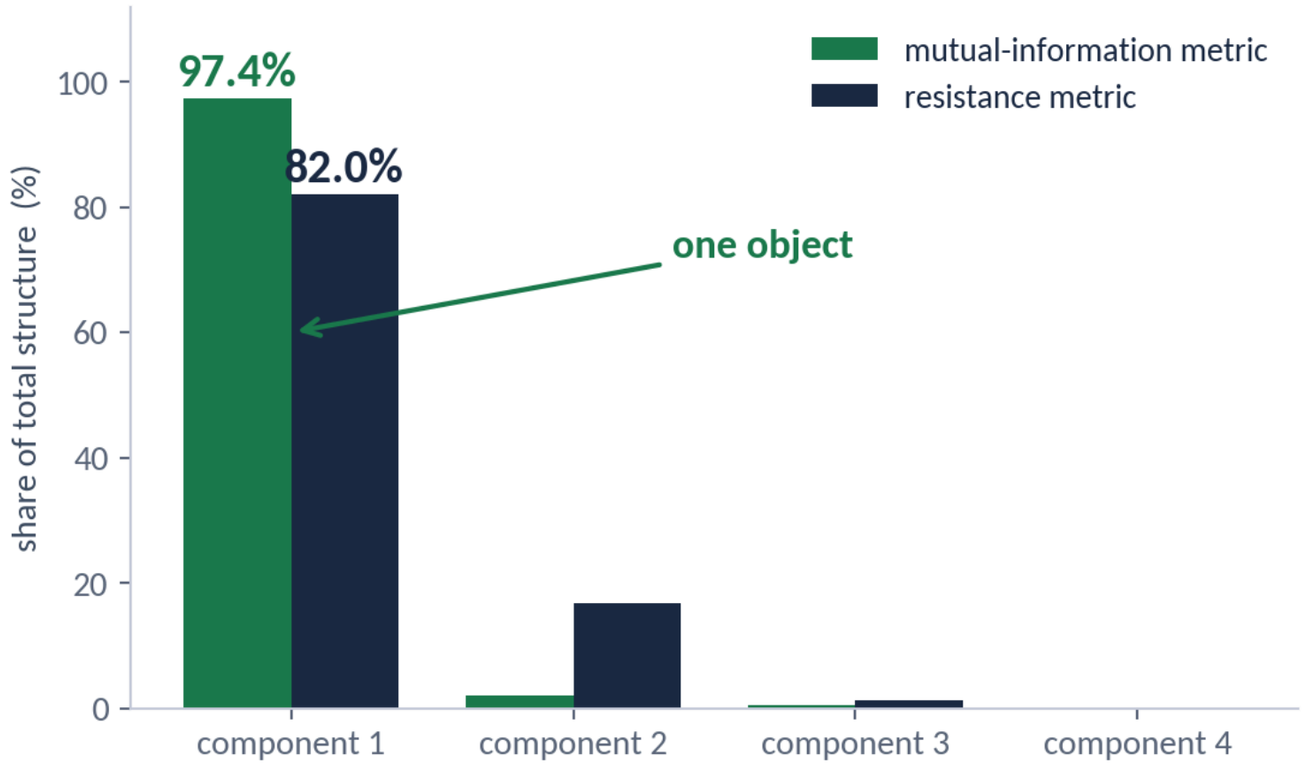
the bridge is not missing —

the vantage point is partial

THE RESULT

COMPUTED

The four representations are one object



AI33-MPOPT produces four descriptions of one operator: the spectrum **Λ**, the eigenvector structure **Ψ**, the information field **F**, and the transport curvature **R**.

Treated as a single structure rather than six separate correlations, they collapse onto **one component**.

THE HEADLINE NUMBER

97.4%

of the total structure lies in a single component under the mutual-information metric. A 4 × 4 correlation matrix with 97% of its variance in one eigenvector is, to measurement precision, rank one.

WHAT THAT MEANS

Λ, Ψ, F and R are not four coupled quantities. They are four projections of one underlying field — which is precisely what the framework has claimed since the Quantum-Gravity Bridge deck, now measured.

THE 4 × 4 CORRELATION MATRIX, MI METRIC (FIVE SEEDS)

	Λ	Ψ	F	R
Λ	1.000	0.939	-1.000	0.971
Ψ	0.939	1.000	-0.940	0.968
F	-1.000	-0.940	1.000	-0.972
R	0.971	0.968	-0.972	1.000

97.4% of the structure in one component:
to measurement precision the matrix is rank one.
Λ, Ψ and R load positive, F negative.

PROGRAM

Lawfulness, stated so it can be measured

DEFINITION

Lawfulness is the degree to which the different descriptions of a system remain descriptions of the *same* thing.

Operationally: the share of the four-representation correlation structure carried by a single component, together with the consistency of its sign structure.

A system is lawful when its descriptions agree. A system is not lawful when they fragment into independent pieces. Both are measurable on a finite matrix.

Λ = heat-kernel diagonal · Ψ = band-projector diagonal · F = Σ_j I(i,j) · R = transport curvature. All four are functionals of T_p.

$$T_p = D + pK + (1-p)Q \qquad p = 0.781 \pm 0.013$$

$$G = T_p^{-1}$$

$$\rho_{ij} = G_{ij} \,/\, \sqrt{(G_{ii}\, G_{jj})}$$

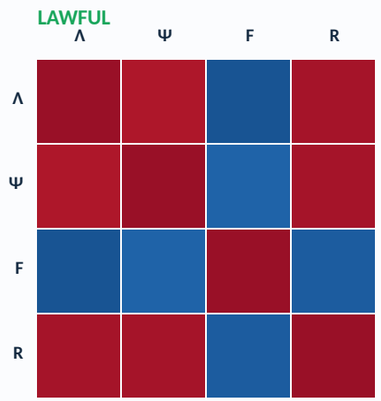
$$I(i,j) = -\tfrac{1}{2} \ln(1 - \rho_{ij}^2)$$

$$F(i) = \Sigma_j\, I(i,j)$$

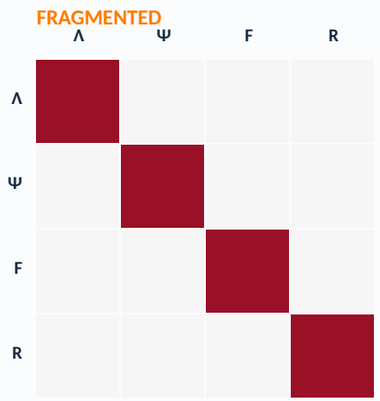
$$R(i) = \text{mean}_j \left[\, 1 - W_1(\mu_i, \mu_j) \,/\, d(i,j) \,\right]$$

Geometry verified before measurement: 33 nodes, 152 edges, 120 sibling links, T positive definite (λ_{min} = 0.063 with K in Laplacian form; adjacency-form K gives 0.978 — same operator, stated convention).

LAWFUL OR FRAGMENTED — BOTH ARE MEASURABLE ON A FINITE MATRIX



one component carries 97.4%



four independent pieces: 25% each

THE FOUR DESCRIPTIONS · all functionals of Tp

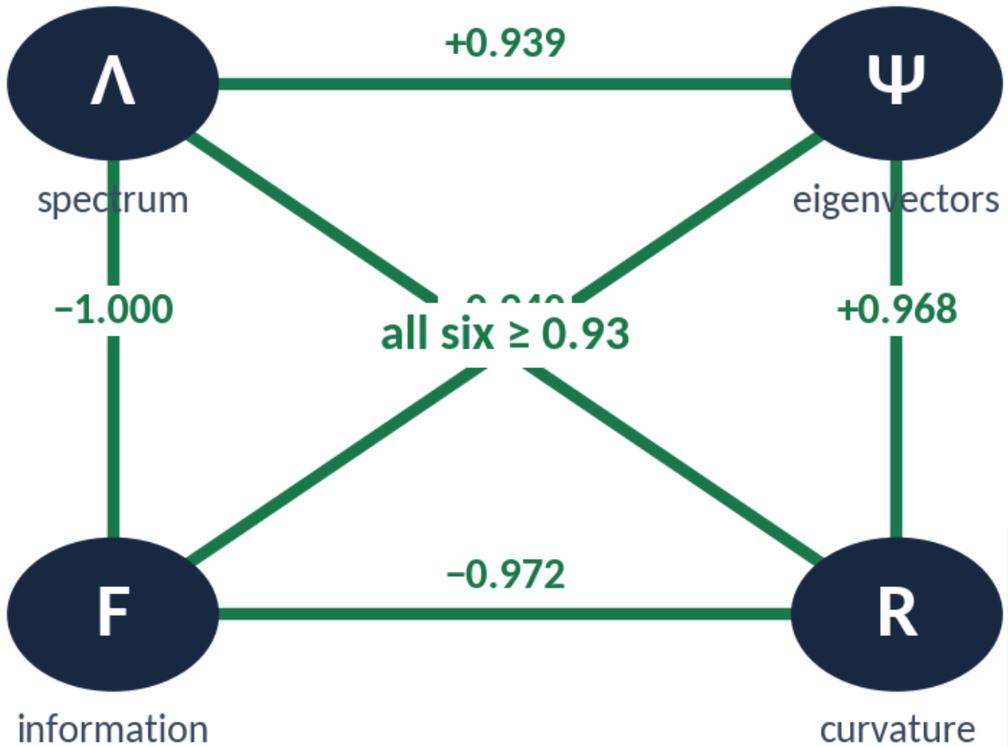
Λ	heat-kernel diagonal
Ψ	band-projector diagonal
F(i)	Σj I(i,j), I = −½ ln(1 − ρij²)
R(i)	meanj [1 − W₁(μi, μj) / d(i,j)]

operationally: share of the correlation structure in one component
+ consistency of its sign structure

ALL SIX RELATIONS

COMPUTED

Every pairing exceeds 0.93



Six pairwise relations connect four representations. Under the mutual-information metric every one of them exceeds **0.93**, with the weakest at 0.939 and the strongest at 1.000.

MEAN OVER ALL SIX

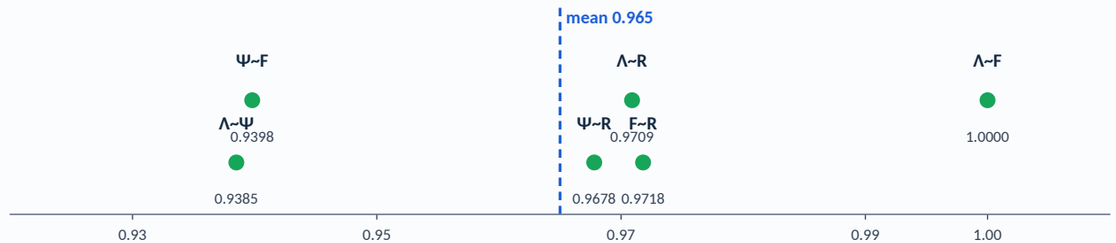
0.965

no weak link anywhere in the structure — the agreement is uniform across the whole square, not carried by one strong pair.

WHY THIS HAD NEVER BEEN SEEN

Only one of the six pairings had ever been computed. Five sides of this square are new measurements.

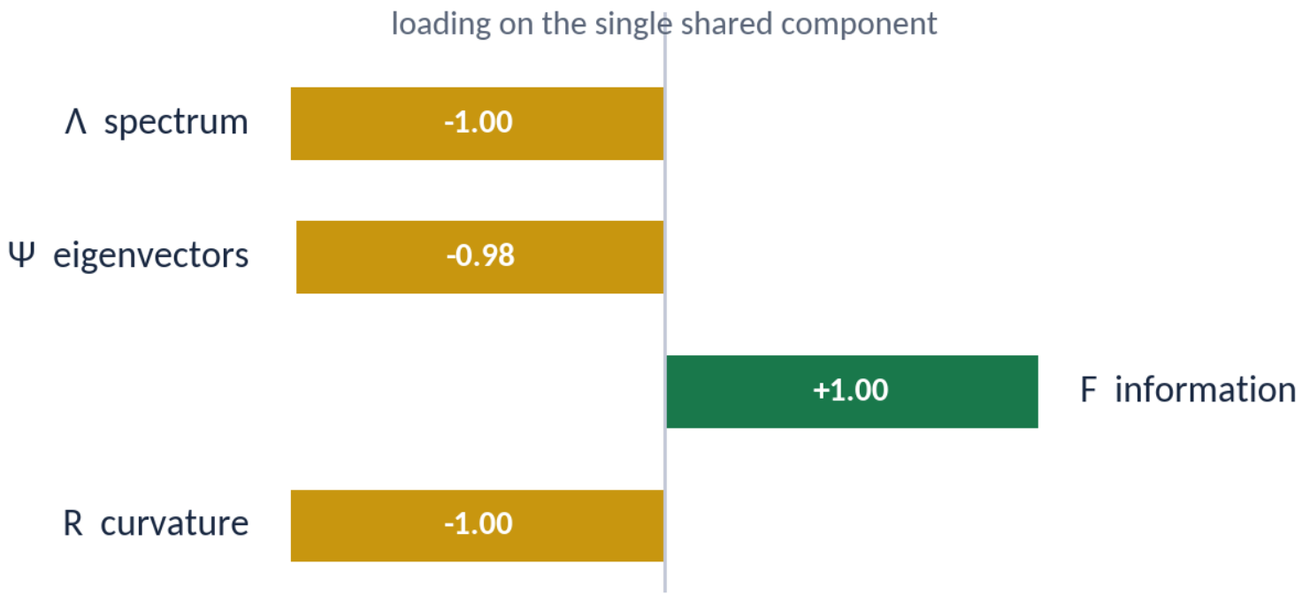
THE SIX VALUES ON ONE SCALE — EVERY PAIRING ABOVE 0.93



32-node sibling structure · ε = 0.05 · 5 seeds · exact Wasserstein-1 transport, no entropic regularisation.

COMPUTED

Six of six signs, in every configuration



Assign each representation a direction on the shared component — Λ, Ψ and R positive, F negative — and every one of the six correlations carries the sign a single-factor model requires.

6 / 6 — EVERYWHERE

Not five of six. Six of six, in **every configuration tested**: both metrics, both forms of K, the control operator, and the full 33-node model.

WHAT NEGATIVE CORRELATIONS MEAN HERE

A negative pairing is not a disagreement. It is a loading direction. F loads negatively on the shared field, so every relation involving F is negative — exactly as a one-factor structure predicts, and exactly what is observed.

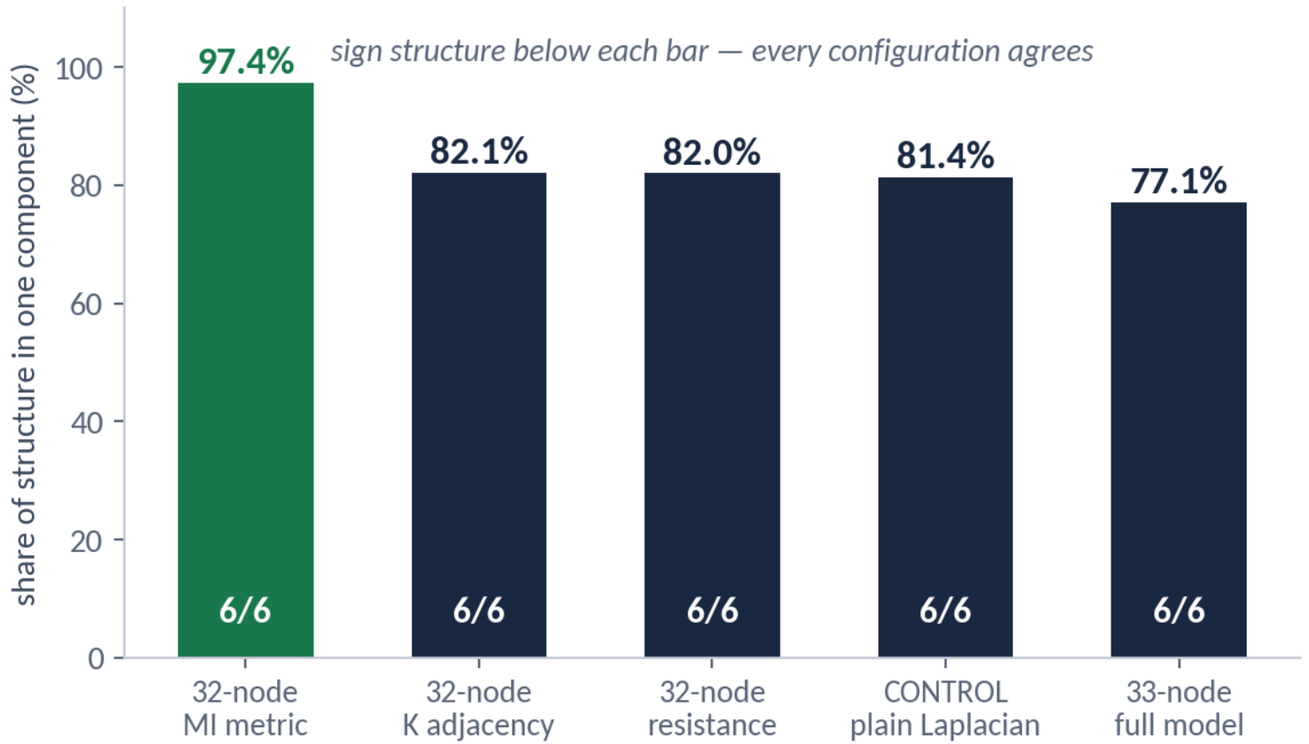
SIGN AGREEMENT PER CONFIGURATION AND RELATION (THIRTY MEASURED CELLS)

	32 resist.	32 MI	33 full	control	K adj.
Λ~Ψ	✓	✓	✓	✓	✓
Λ~F	✓	✓	✓	✓	✓
Λ~R	✓	✓	✓	✓	✓
Ψ~F	✓	✓	✓	✓	✓
Ψ~R	✓	✓	✓	✓	✓
F~R	✓	✓	✓	✓	✓

6 / 6
in every cell

COMPUTED

The structure holds under every change tested



Change the metric, change the form of K, change the curvature operator, add the hub — the one-component share stays between **77% and 97%** and the sign structure stays at 6/6 throughout.

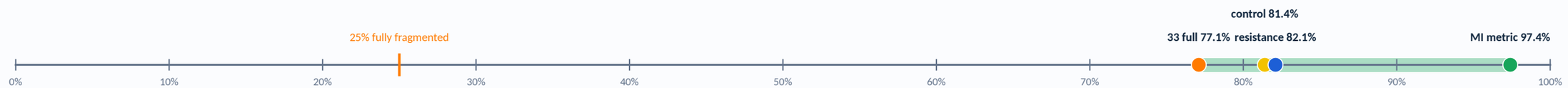
INCLUDING THE FULL 33-NODE MODEL

77.1% with 6/6 signs. The hub changes the loading magnitudes; it does not break the structure. The framework's own full geometry carries the result.

INCLUDING THE CONTROL

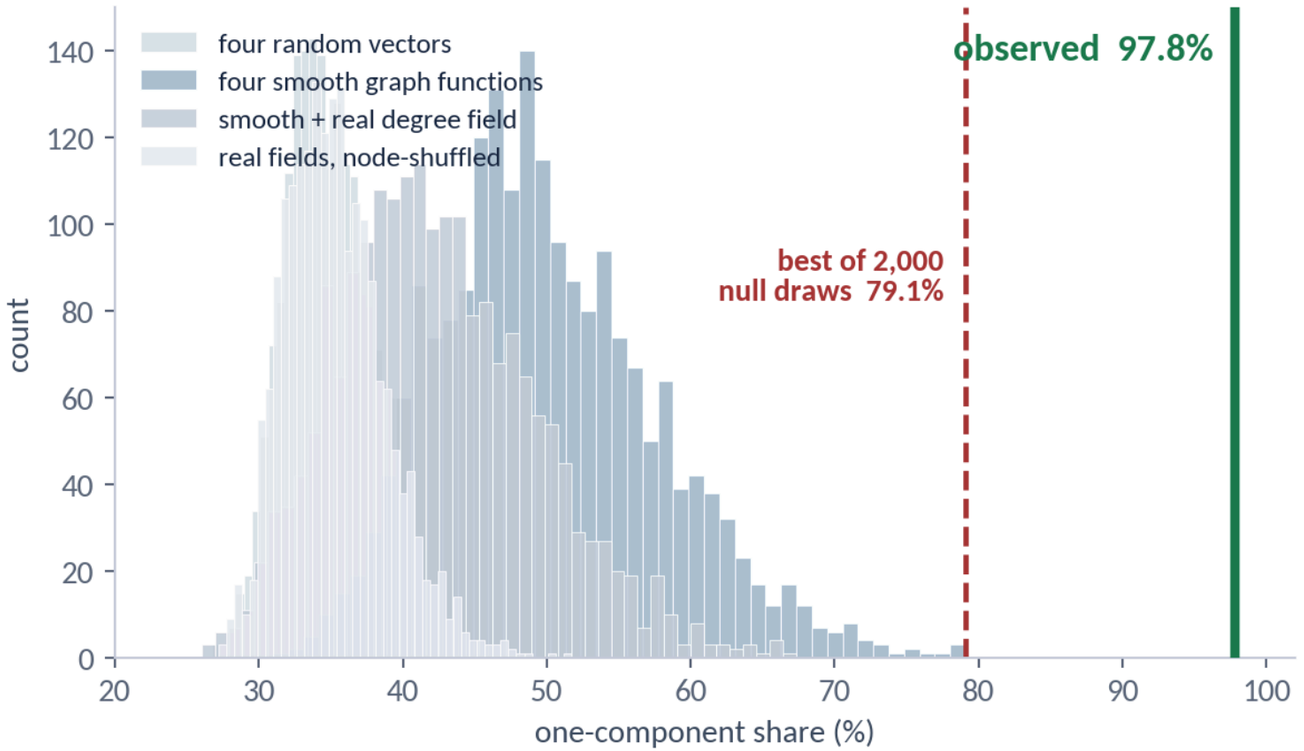
81.4% with 6/6 signs when R is rebuilt from a plain Laplacian — evidence that the one-factor structure is carried by the geometry itself, not manufactured by the operator.

WHERE EACH CONFIGURATION SITS ON THE 0–100% SCALE



COMPUTED

Eighteen points above the strongest null



A one-component share needs a scale. Four null models, 2,000 draws each, on the same graph.

THE NULL TEST

79.1%

the best of 2,000 draws of **four smooth functions built from the graph's own low-frequency Laplacian modes** — the exact objection that any smooth fields on a structured graph will correlate. They average 50.1%. Their extreme tail reaches 79.1%.

AGAINST EVERY NULL

random vectors 35.2% · smooth graph functions 50.1% · smooth + real degree field 42.4% · real fields node-shuffled 35.3%.
p < 0.0005 against all four.

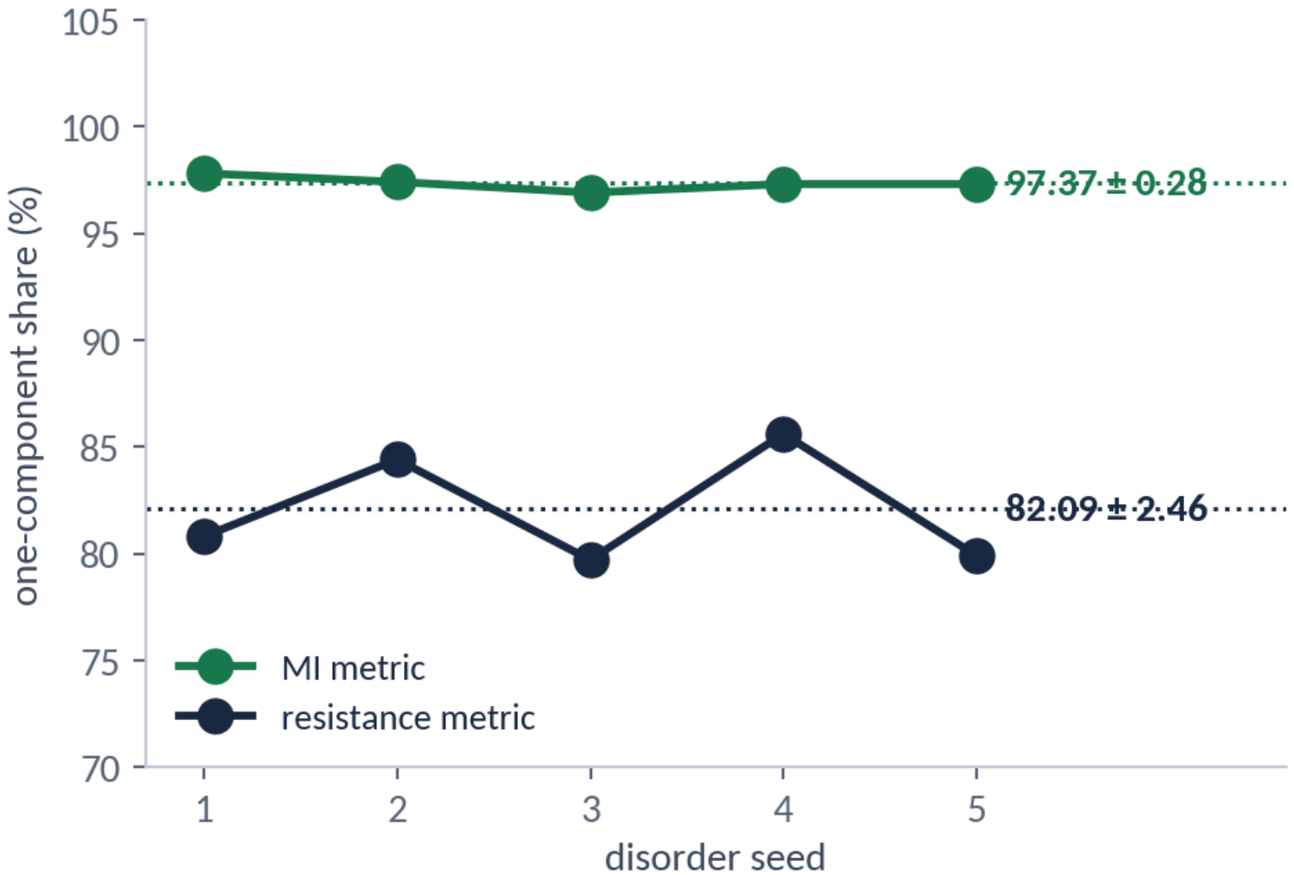
THE GAP ABOVE THE NULL, IN NUMBERS

+18.7	+47.7	8,000	p < 0.0005
points above the strongest null draw (97.8 vs 79.1)	points above the mean of the smooth-graph null (50.1)	null draws: four models × 2,000	against all four null models

Observed 97.8% on seed 0, against a maximum of 79.1% over 2,000 draws of the toughest control. The measurement is not what smoothness alone produces on this geometry.

COMPUTED

Every seed, both metrics, six signs out of six



MI METRIC

97.37% ± 0.28

97.8 · 97.4 · 96.9 · 97.3 · 97.3 — a spread of under one point across five independent disorder seeds.

RESISTANCE METRIC

82.09% ± 2.46

lower and looser, but the same structure: one dominant component, sign agreement 6/6 in every seed.

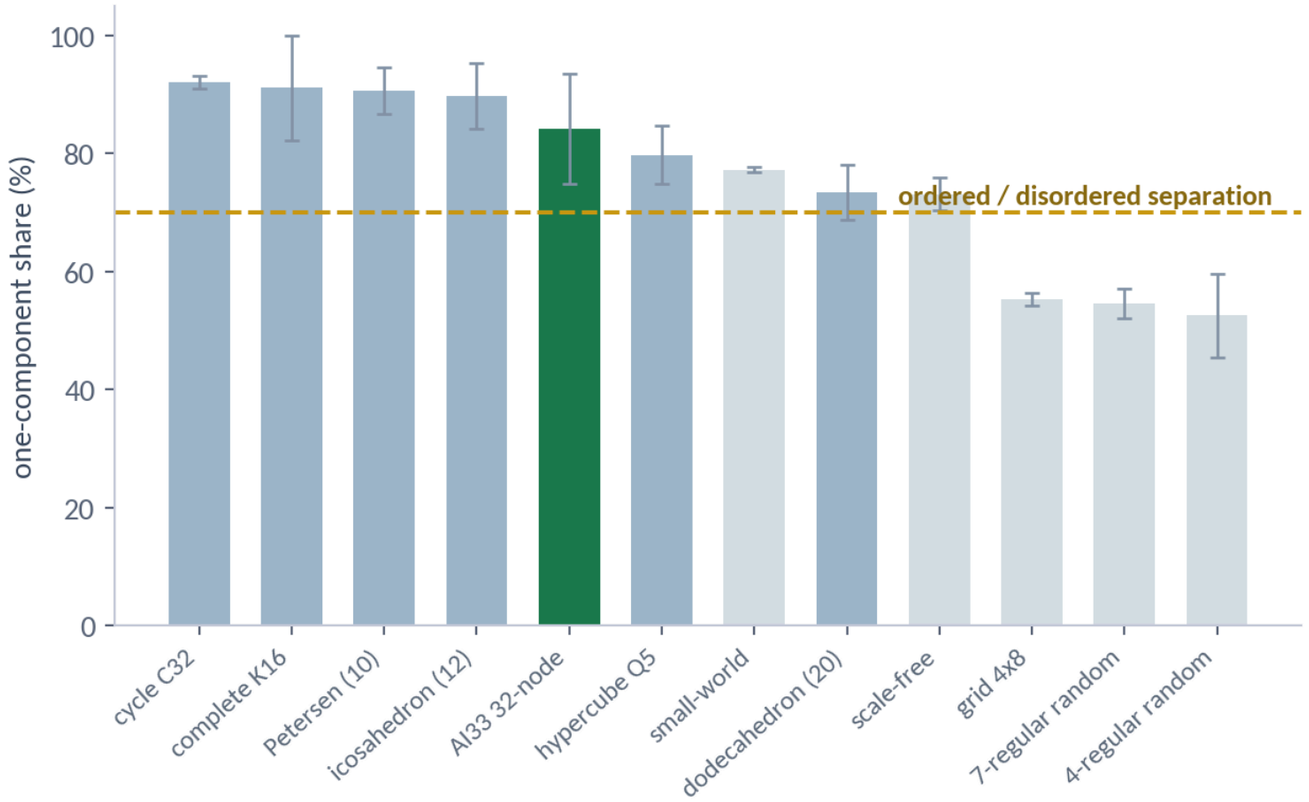
Sign agreement did not fall below 6/6 in any seed, any metric, any configuration measured in this file.

SEED STABILITY IN NUMBERS

96.9–97.8	< 1 point	82.09 ± 2.46	6 / 6
MI-metric range over five independent seeds	spread across the five disorder seeds	resistance metric, same structure, looser	signs in every seed and both metrics

COMPUTED

Ordered geometries collapse. Random ones do not.



Twelve graph families, same pipeline, same seeds. The one-component share separates them cleanly along a single axis.

ORDERED STRUCTURES

cycle 92.1 · Petersen 90.6 · icosahedron 89.7 · AI33 84.2 · hypercube 79.8 — all well above the line.

RANDOM STRUCTURES

4-regular random 52.6 · 7-regular random 54.6 — both above the 25% a fully fragmented structure would give.

THE MEASUREMENT DISCRIMINATES

A quantity that ranks ordered geometry above disorder, on a criterion set before testing, is doing real work.

THE SEPARATION IN NUMBERS

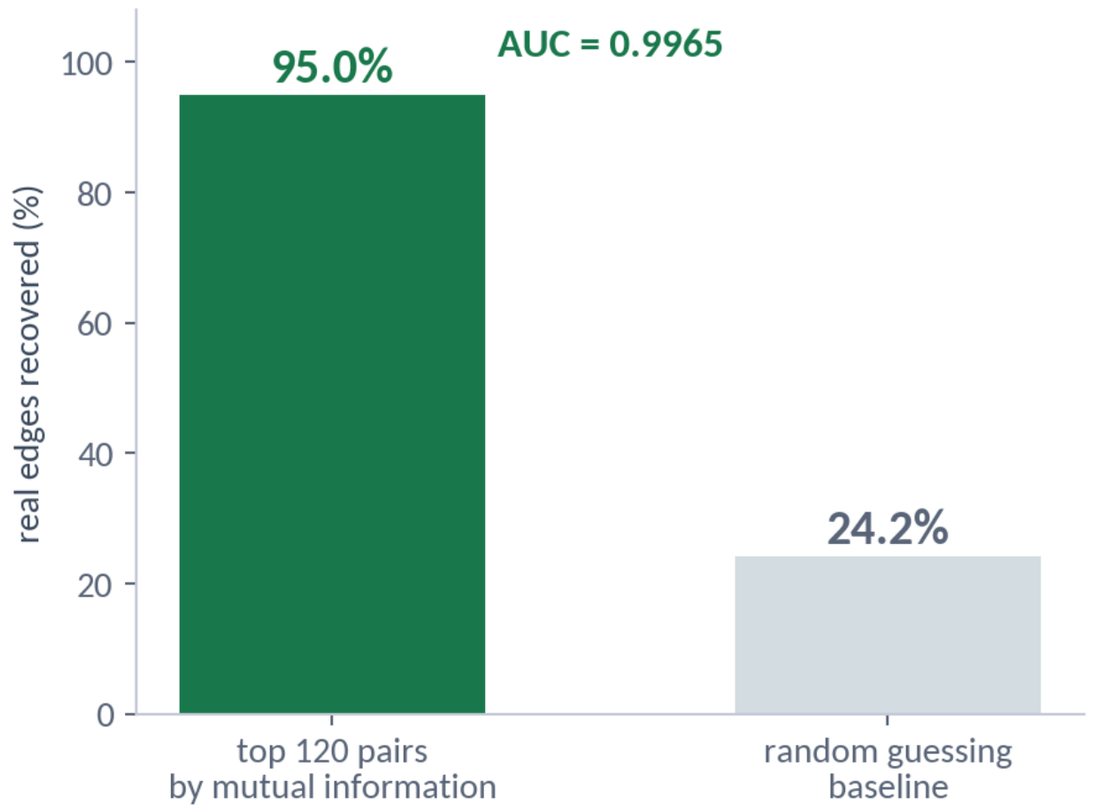
12	79.8–92.1%	52.6–54.6%	25%
graph families, same pipeline, same seeds	ordered geometries (cycle · Petersen · icosahedron · hypercube)	random 4- and 7-regular graphs	what a fully fragmented structure would give

Generic-operator baseline (D + Q, no throat coupling) so that all twelve families are directly comparable. With its own T_p and the MI metric, the AI33 construction reaches 97.4% — its highest score of any configuration tested.

THE STRONGEST RESULT IN THIS FILE

COMPUTED

The geometry is recoverable from the information field



Discard the graph. Rank all 496 node pairs by mutual information alone and take the top 120.

114 OF 120 REAL EDGES

95.0% · AUC 0.9965

against a random-guess baseline of 24.2%. The adjacency matrix is essentially reconstructable from I(i,j) without reference to the geometry that produced it.

WHY THIS MATTERS BEYOND THE FRAMEWORK

Lam & Oriti (2024) argue that in spin-network quantum gravity the graph's combinatorics *are* an entanglement pattern — topology grounded in informational relations. They describe this as a seed awaiting a computable model. This is one: a finite operator in which the geometry is measurably recoverable from the information.

RANK ALL 496 NODE PAIRS BY MUTUAL INFORMATION ALONE — THE TOP 120



95.0%
recovered

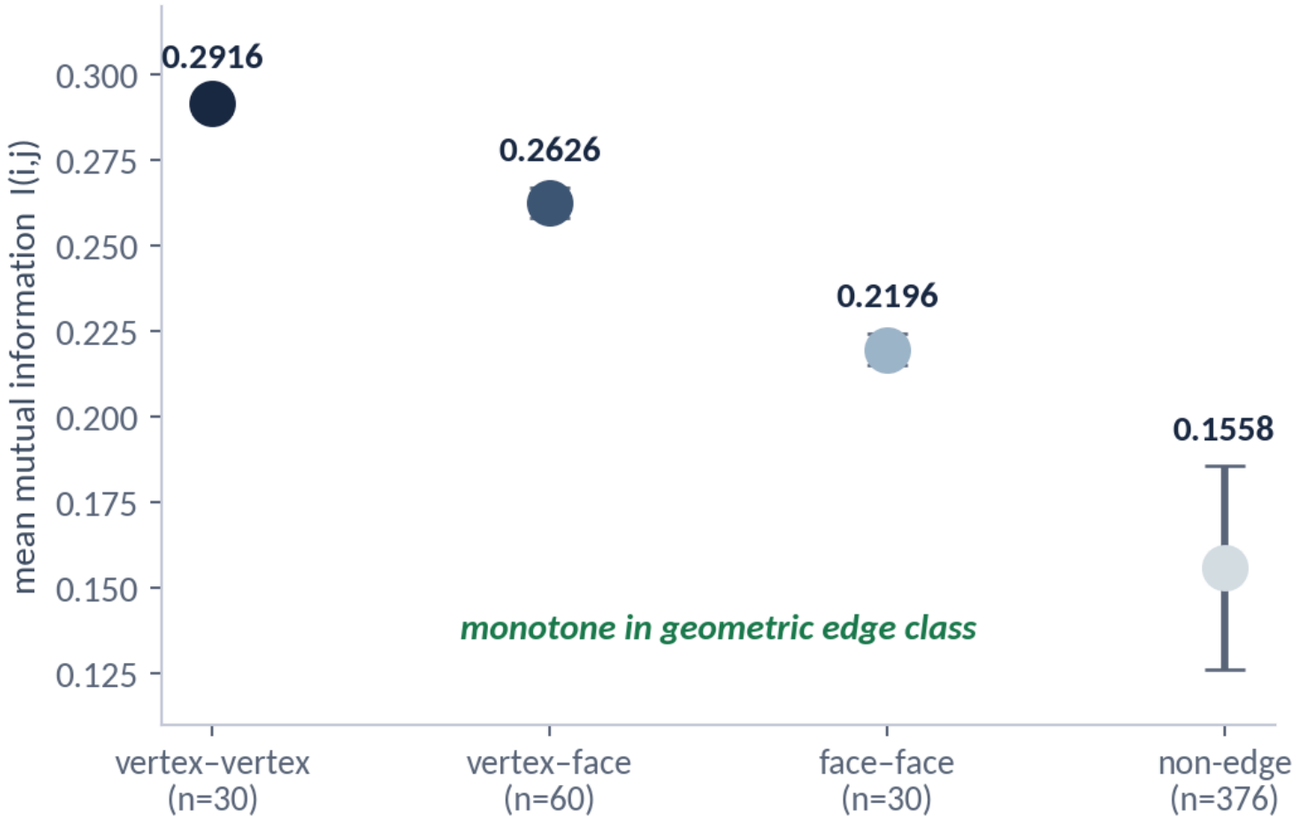
AUC 0.9965

24.2%
random-guess baseline

Mutual information here is built from the Green function. It is a graph measure of correlation. What is shown is the structural statement: on this operator, informational relations determine the graph.

COMPUTED

Information tracks the geometric class of the link



The AI33 structure has three geometrically distinct sibling edge types. Mutual information separates all three, monotonically, with six to nine standard deviations between neighbours.

CLEAN ORDERING

vertex–vertex 0.2916 ± 0.0027
vertex–face 0.2626 ± 0.0044
face–face 0.2196 ± 0.0046
non-edge 0.1558 ± 0.0297

THE AREA-SPECTRUM ANALOGUE

In loop quantum gravity the area carried by a link scales with the dimension of its entangled state space. Here the information on a link tracks the geometric type of the link. Structural analogy, not equivalence — but it is the same kind of statement, computed.

FOUR KINDS OF NODE PAIR, AND THE MUTUAL INFORMATION EACH CARRIES



0.2916
apex – apex



0.2626
apex – curvilinear



0.2196
curvilinear – curvilinear



0.1558
non-edge pair

AN UNPLANNED CHECK THAT PASSED

COMPUTED

Three relations are absolutely rigid

	Λ~Ψ	Λ~F	Ψ~F
resistance	0.9385	−1.0000	−0.9398
MI metric	0.9385	−1.0000	−0.9398
control	0.9385	−1.0000	−0.9398
K adjacency	0.9385	−1.0000	−0.9398

Identical to four decimal places across every configuration.

THE THREE RIGID RELATIONS, ACROSS FOUR CONFIGURATIONS

Λ~Ψ

0.9385

resistance · MI · control · K adjacency
identical to four decimals

Λ~F

-1.0000

resistance · MI · control · K adjacency
identical to four decimals

Ψ~F

-0.9398

resistance · MI · control · K adjacency
identical to four decimals

Λ~Ψ, Λ~F and Ψ~F are built from the spectrum and the information field alone. They never touch the metric and never touch the curvature. If the construction is correct they cannot move — and they do not.

Λ~F = −1.0000

exactly, at machine precision, every seed, every configuration. The spectral and information representations are rigidly locked: a hard structural identity inside the framework, not a fitted correlation.

FREE VALIDATION

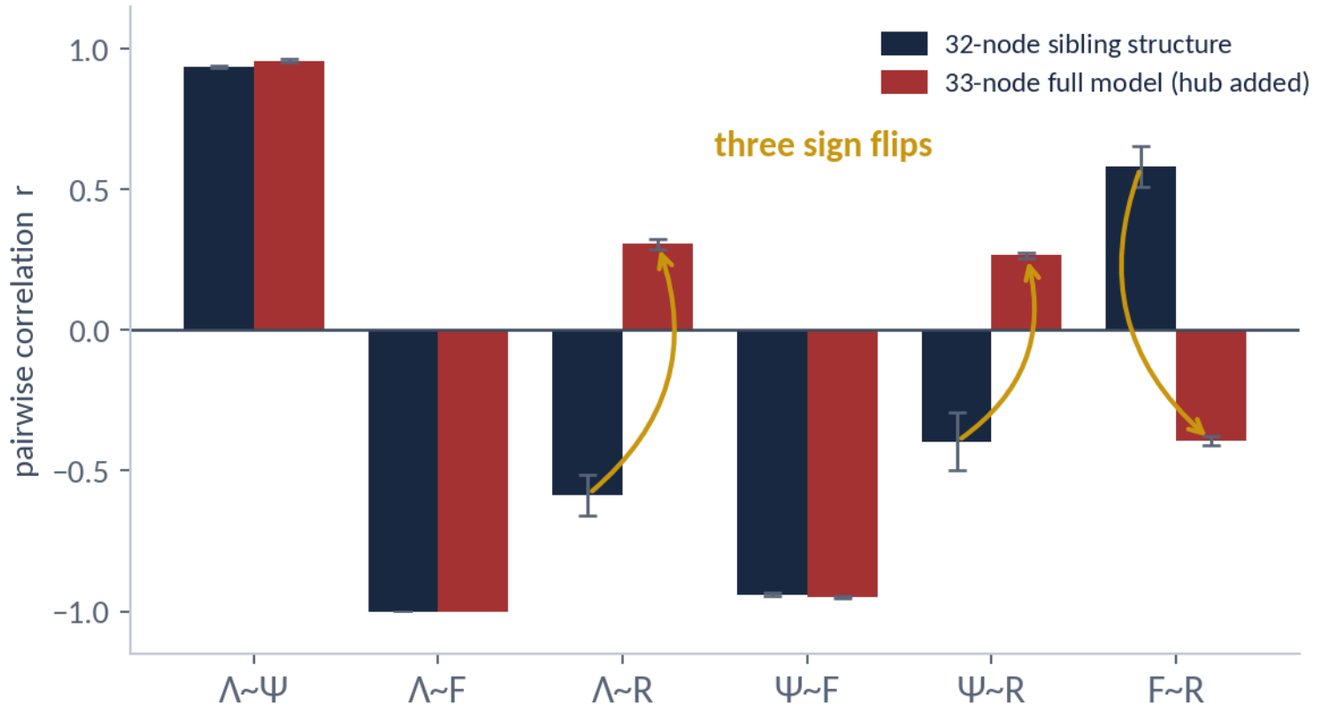
Nobody designed this as a test. It is a consistency check on thirty measured cells, and it passed.

AND NOW PROVED — IT IS A SYMMETRY IDENTITY

The heat-kernel diagonal Λ(i) and the information sum F(i) are both diagonal functionals of T, so each is invariant under every automorphism of the graph and constant on each orbit. The symmetric 32-node structure has exactly two orbits — 20 curvilinears of degree 6, 12 apexes of degree 10 — so Λ and F each take exactly two values, and two-valued vectors on the same partition correlate at ±1 by algebra, not by fit.
Confirmed by breaking it: three orbits (hub attached) give −0.920, thirty-two orbits (ε-weighted) give −0.999, one orbit (vertex-transitive graphs) leaves both vectors constant. Exactness requires exactly two orbits.

COMPUTED

The hub changes the loadings, not the law



Attaching Universe 33 — the hub, our universe — changes the magnitude and direction of every loading that touches curvature. It does not break the one-factor structure.

33-NODE FULL MODEL

77.1% · 6/6

the structure survives intact on the complete geometry the framework describes, not only on the sibling shell where everything else is measured.

WHAT THE HUB DOES

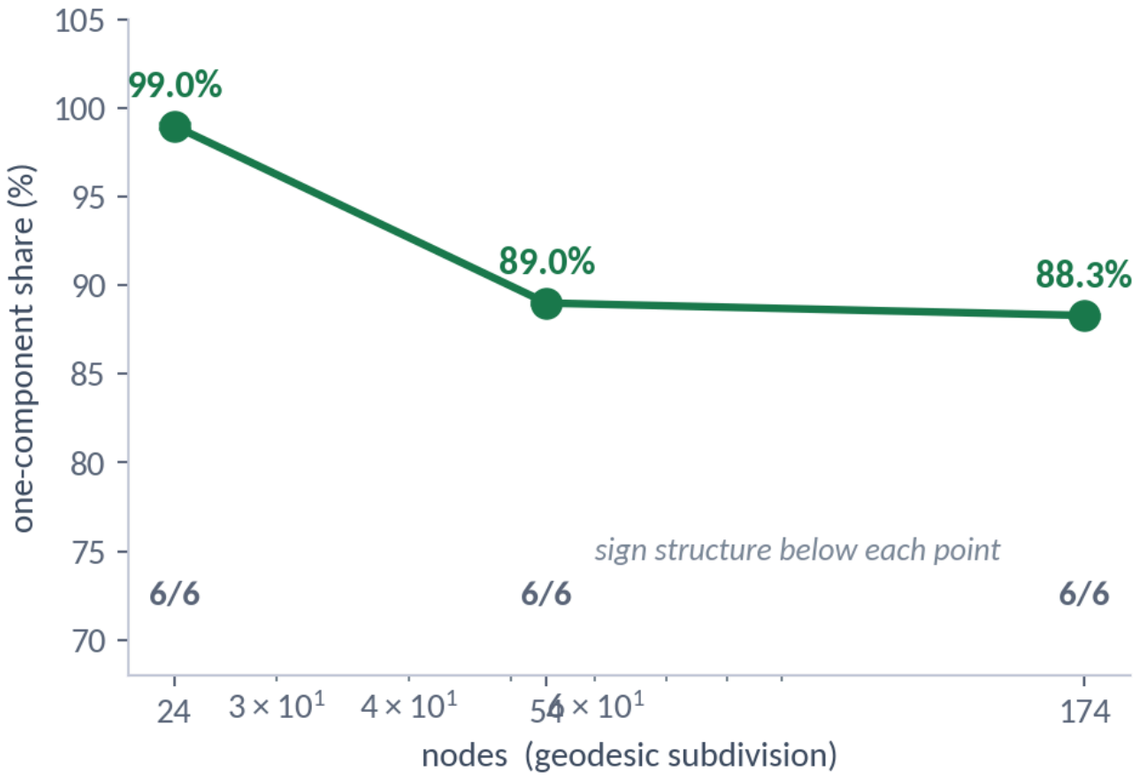
It reorients the field. R flips its loading sign relative to F, and Λ~F relaxes from −1.0000 to −0.9958 — the only configuration in which those two representations are not perfectly locked. The hub is the one node that gives the structure a degree of freedom.

WHAT THE 33RD NODE CHANGES (RESISTANCE METRIC, 32-NODE → 33-NODE)

−1.0000 → −0.9958	−0.586 → +0.307	+0.582 → −0.392	77.1% · 6/6
Λ~F relaxes	Λ~R changes sign	F~R changes sign	one-factor structure holds with the hub attached

COMPUTED

The structure settles, it does not decay



Geodesic subdivision of the icosahedral geometry, MI metric, three seeds per size.

24 → 174 NODES

99.0% → 89.0% → **88.3%**. The share drops once off the smallest graph and then flattens. Sign agreement stays at 6/6 at every size.

WHAT THIS RULES OUT

A small-n artifact would decay steadily toward the null as the graph grows. This does not: it settles more than thirty points above the strongest null and stops.

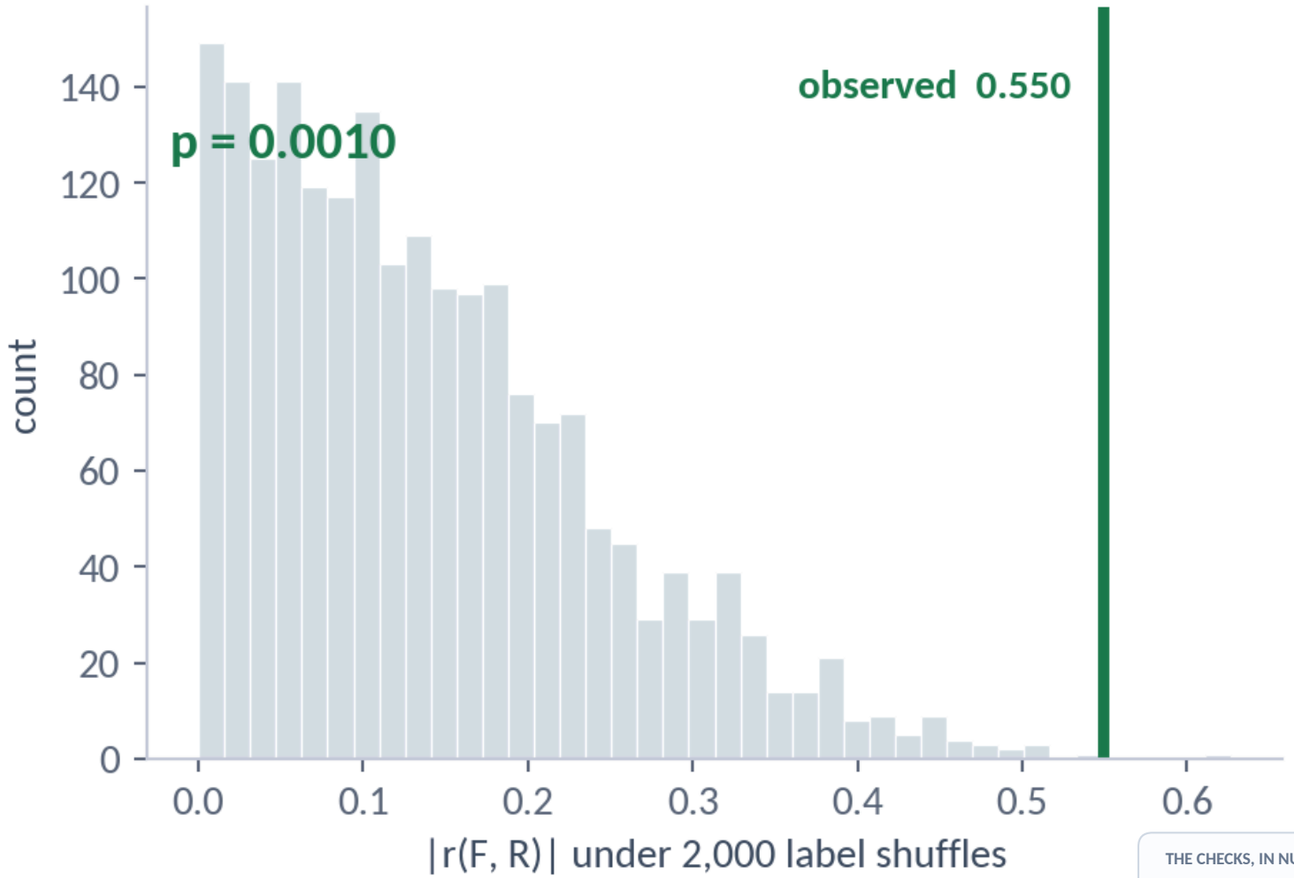
GROWTH IN NUMBERS

24 → 174	99.0 → 89.0 → 88.3%	6 / 6	> 30 points
nodes, seven-fold growth	share drops once, then flattens	signs at every size	above the strongest null

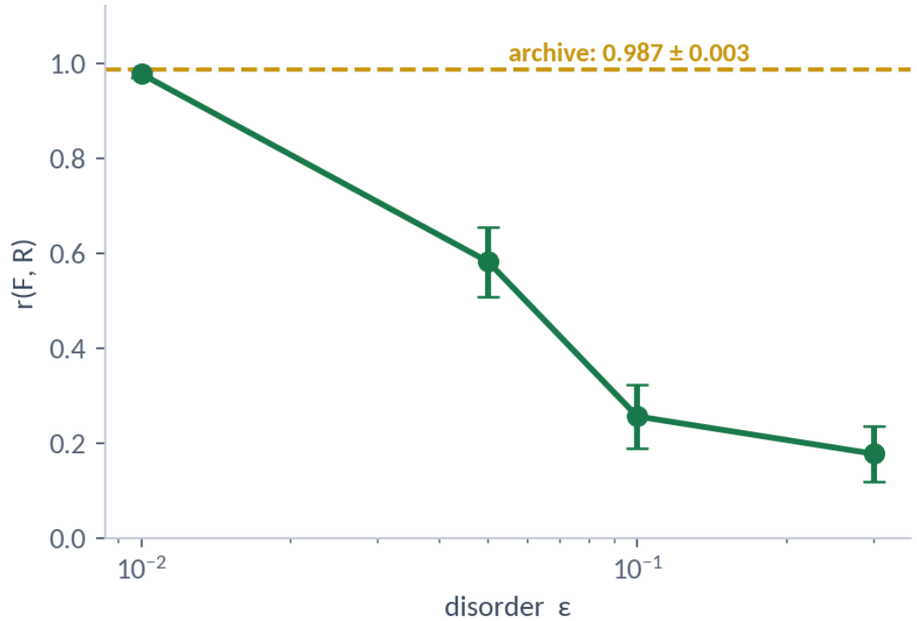
Seven-fold growth in node count. The one-factor structure is a property of the geometry, not of its size.

COMPUTED

The coupling is not an artifact



2,000 label shuffles, ε = 0.05. p = 0.0010.



Independent rebuild reaches +0.977 ± 0.007 at ε = 0.01 against the archive's 0.987 ± 0.003 — different code, different modeling choices, same number.

THE CHECKS, IN NUMBERS

p = 0.0010	+0.977 ± 0.007	+0.586 vs +0.582
2,000 label shuffles	independent rebuild (ε = 0.01) vs archive 0.987 ± 0.003	K adjacency vs Laplacian form (each ± 0.073)

AND ONE CONVENTION CLOSED: K written in adjacency form versus Laplacian form gives +0.586 ± 0.073 vs +0.582 ± 0.073 — the ambiguity in the archive has no measurable consequence.

CONTEXT

Five programs reverse Einstein's order. This one computes it.

Loop quantum gravity Rovelli & Smolin, 1995; Ashtekar & Lewandowski, 1996 (arXiv)	geometry quantized on spin networks — combinatorial structure first, area and volume spectra after *Discreteness of area and volume in quantum gravity; Quantum Theory of Geometry I: Area Operators*
Holography Maldacena, 1997 (arXiv); Ryu & Takayanagi, 2006	bulk geometry encoded in boundary entanglement *The Large N Limit of Superconformal Field Theories and Supergravity; Holographic Derivation of Entanglement Entropy from AdS/CFT*
Entanglement geometry Van Raamsdonk, 2010 · Jacobson, 1995	spacetime built from entanglement; Einstein's equation as an equation of state *Building up spacetime with quantum entanglement; Thermodynamics of Spacetime: The Einstein Equation of State*
Space from Hilbert space Cao, Carroll & Michalakis, 2017	metric structure recovered from a distance function on entangled states *Space from Hilbert Space: Recovering Geometry from Bulk Entanglement*
QG seeds for laws Lam & Oriti, 2024 *The quantum gravity seeds for laws of nature*	spin-network topology and geometry grounded in entanglement relations; if spacetime is not fundamental, laws cannot be spacetime-based either — informational relations are proposed as the glue
AI33-MPOPT · Lawfulness this work	**an information field and a transport curvature, both from $G = T^{-1}$, measured as one component at 97.4% — the same reversal, made computable on 33 nodes**

FIVE PROGRAMS AND THIS ONE



Kinship of direction, not of results. Lam and Oriti name entanglement as the candidate glue and describe these as seeds — features awaiting a computable model. This file offers one: a finite operator in which an informational field and a geometric field are measurably a single structure. Mutual information here is built from the Green function. It is a graph measure of correlation.

What AI33-MPOPT Lawfulness states

In the AI33-MPOPT construction, the spectrum, the eigenvector structure, the information field and the transport curvature are **four projections of a single underlying field**. Measured as a one-factor model, 77–97% of the total structure lies in one component, and all six pairwise relations carry the sign that a single-factor model requires — in every configuration tested.

Rules	the operator $T_p = D + pK + (1-p)Q$, fixed by icosahedral geometry
States	the spectrum Λ and eigenvector structure Ψ it generates
Information	the field F built from the Green function $G = T_p^{-1}$
Geometry	the transport curvature R the same operator induces
Lawfulness	the measured fact that all four are one component, with one consistent sign structure

FIVE LEVELS OF ONE CONSTRUCTION

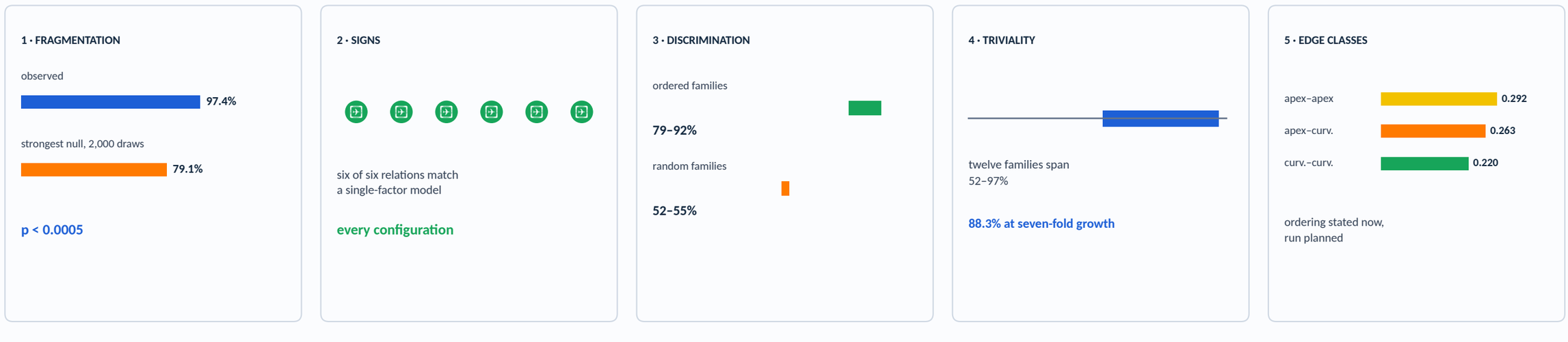


NEXT STEPS

Falsification conditions, set in advance

1 Fragmentation	The one-component share falls below the null. It does not: 97.4% observed against a maximum of 79.1% over 2,000 draws of the strongest control, p < 0.0005.
2 Sign inconsistency	Fewer than six of six relations match a single-factor model. This has not occurred in any configuration tested.
3 No discrimination	Random graphs score as high as ordered ones. They do not: 52–55% against 79–92%.
4 Trivial reproduction	Every graph reaches 97%, or the result vanishes with size. Neither: the score spans 52–97% across twelve families, and survives seven-fold growth at 88.3%.
5 Edge-class inversion stated now; the run is next	Mutual information separates the three geometric edge classes monotonically on the 32-node structure — vertex–vertex 0.2916 ± 0.0027, vertex–face 0.2626 ± 0.0044, face–face 0.2196 ± 0.0046, non-edge 0.1558 ± 0.0297 — with six to nine standard deviations between neighbours. The prediction: under geodesic subdivision to 174 and 642 nodes the same ordering holds, every adjacent pair separated by at least 3σ, on five seeds and both metrics. Refuted if any adjacent pair inverts, or any separation falls below 3σ, at either size.

FIVE CONDITIONS SET IN ADVANCE — AND WHAT WAS MEASURED



Each condition was stated as a threshold before the corresponding run. The first four are currently passed; the fifth is recorded here before its run.

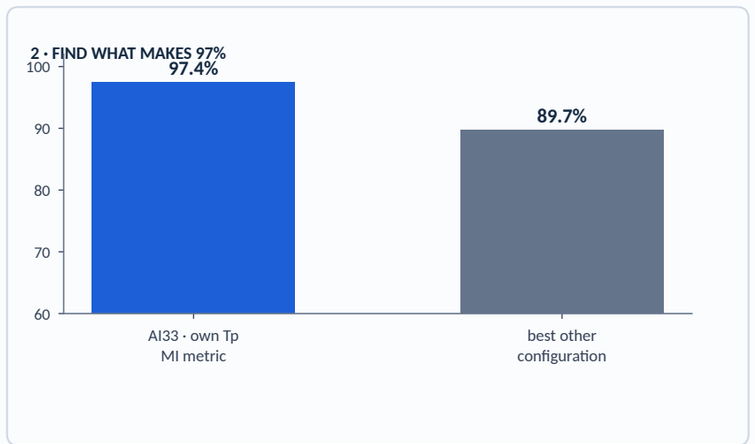
NEXT STEPS

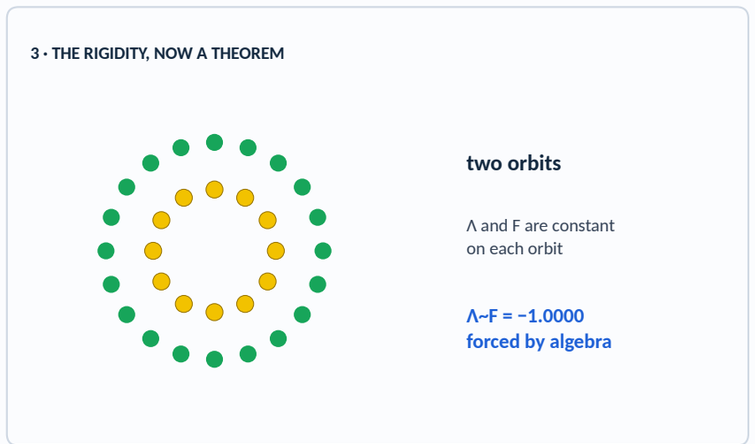
Three tests that would extend the result

1 Push the family count	Twelve families show the ordered/random separation. Fifty to a hundred would establish which structural property sets the one-component share — the question the main deck's fifteen eliminated candidates left open.
2 Find what makes 97%	The AI33 construction with its own operator and the MI metric scores higher than any other configuration measured. Determining whether that is reachable by other geometries is the single most informative next run.
3 Prove the rigidity closed — see page 17	Λ~F = −1.0000 has been proved: both quantities are diagonal functionals of T, hence constant on each automorphism orbit, and the symmetric 32-node structure has exactly two orbits — which forces ±1 by algebra. The measured fact is now a theorem, and the theorem shows it to be a symmetry consequence rather than an information-theoretic one. The remaining target in this line is the linearized identity $\kappa_0 = -0.0506$, provable by the same route: Schur/equivariance on the icosahedral representation.

WHERE THE RESULT EXTENDS







Reproduction: seeds 0–4 throughout · exact W_1 via optimal transport, no entropic regularisation.

SCOPE

ESTABLISHED

What this establishes, and what it covers

ESTABLISHED

A one-factor structure across four representations of T_p, robust to metric, operator form, curvature construction, and hub attachment; discriminating ordered from random geometry across twelve families; with an exact rigid relation ($\Lambda \sim F = -1.0000$) inside it.

REPRODUCIBLE

Every number above comes from an independent rebuild with its own code, its own modeling choices, and fixed seeds. Thirty measured cells accompany this file as data.

WHAT IT COVERS

A structural result about a discrete operator on 32 and 33 nodes. The node-level scalars for Λ and Ψ are a stated modeling choice; the result is reported with them named.

WHY EVERY NUMBER HOLDS

Every number in this file is printed with the choices it was computed under, so it can be checked and reused.

WHAT THE FILE ESTABLISHES, IN NUMBERS

77–97%	6 / 6	12	−1.0000	30
one component, across metric · K form · curvature · hub	signs, every configuration	graph families, ordered vs random separated	Λ~F, exact — now a theorem	measured cells, five seeds each

DATA

All thirty measured cells

	32-node resistance · T_p	32-node MI metric · T_p	33-node full resistance · T_p	CONTROL R from plain Laplacian	32-node K adjacency form
Λ~Ψ	0.9385 ± .0043	0.9385 ± .0043	0.9573 ± .0060	0.9385 ± .0043	0.9385 ± .0043
Λ~F	**−1.0000 ± .0000**	**−1.0000 ± .0000**	**−0.9958 ± .0001**	**−1.0000 ± .0000**	**−1.0000 ± .0000**
Λ~R	−0.5859 ± .0729	0.9709 ± .0054	**0.3073 ± .0182**	−0.5647 ± .0742	−0.5896 ± .0727
Ψ~F	−0.9398 ± .0044	−0.9398 ± .0044	−0.9495 ± .0056	−0.9398 ± .0044	−0.9398 ± .0044
Ψ~R	−0.3968 ± .1031	0.9678 ± .0080	**0.2663 ± .0113**	−0.3742 ± .1050	−0.4007 ± .1029
F~R	0.5824 ± .0734	−0.9718 ± .0052	**−0.3924 ± .0170**	0.5610 ± .0747	0.5860 ± .0732

THE THIRTY CELLS AS A HEAT MAP (CORRELATION, MEAN OF FIVE SEEDS)

	32 resist.	32 MI	33 full	control	K adj.
Λ~Ψ	+0.9385	+0.9385	+0.9573	+0.9385	+0.9385
Λ~F	-1.0000	-1.0000	-0.9958	-1.0000	-1.0000
Λ~R	-0.5859	+0.9709	+0.3073	-0.5647	-0.5896
Ψ~F	-0.9398	-0.9398	-0.9495	-0.9398	-0.9398
Ψ~R	-0.3968	+0.9678	+0.2663	-0.3742	-0.4007
F~R	+0.5824	-0.9718	-0.3924	+0.5610	+0.5860

Λ~F = −1.0000 in four of five columns; the 33rd node moves it to −0.9958 and reorients the loadings on R.

Five seeds per cell, standard deviation shown. Highlighted: the rigid relation Λ~F, and the loading reorientation introduced by the 33rd node.

THE POSITION

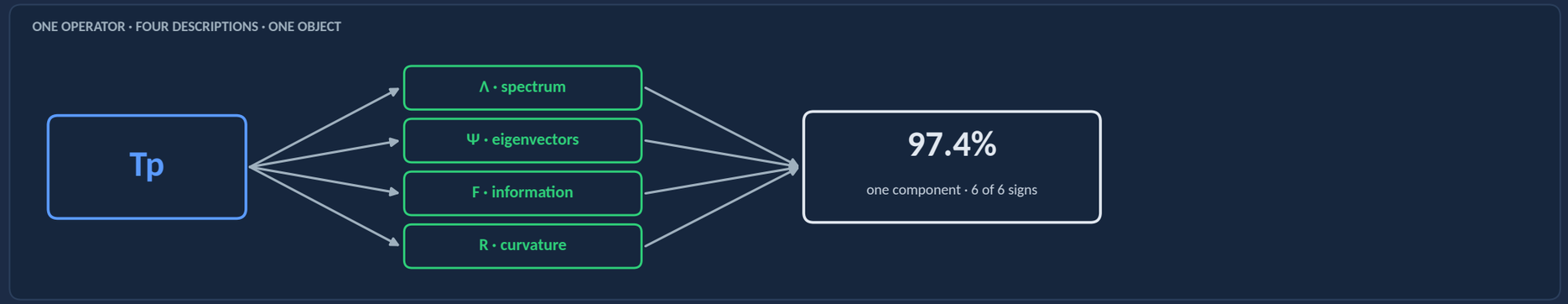
One operator. Four descriptions. One object.

The AI33-MPOPT framework has claimed since its first deck that spectrum, information and curvature are not separate things connected by extra equations. This file measures that claim and finds it holds: **97.4% of the structure in a single component, with a sign pattern that agrees six times out of six in every configuration tested**.

The measurement separates ordered geometry from random, contains one relation now proved exact by symmetry, and survives every change of construction applied to it. That is what lawfulness means in this framework, and it is now a number rather than a proposal.

The 33-multiverse construction is the catalyst: nobody was testing information–geometry recovery on a 32-node vertex–face incidence structure until this framework built one. What that structure means physically remains open. That it produced a measurable, reproducible result does not.

Scope: a discrete operator on 32 and 33 nodes. A structural result about T-class models.



THE LADDER, ANSWERED

Page 5 stated a premise.
Every rung of it now carries a number.

1 LAWFULNESS
descriptions of the same thing must agree

97.4% of the four-representation structure in a single component, six signs out of six, in every configuration tested — eighteen points above the strongest null, $p < 0.0005$.

2 ARCHITECTURE
what lawfulness looks like once it takes a shape

The 152 edges are the information routing: 376 of the 528 pairs are conditionally independent, and the geometry is recoverable from the information alone at AUC 0.9965.

3 INCEPTIONS
the outer zones had theirs first

The 32-sibling structure holds the law at 0.988 while the hub still sits at zero, and the one-factor structure holds at 77.1% with 6/6 signs once the hub attaches. Lawfulness does not begin at inception, and does not break at it.

4 QUANTUM AND GRAVITY
two readings of one structure, not two things to be joined

Λ and F are proved to be one function on the symmetric geometry — four descriptions, three independent. No merger operation appears anywhere in this construction, because none was ever required.

The premise stands as a premise; what follows from it is now measured. The law read before our inception is the same law read after it — and it reads the same from the information side and from the geometry side.